

Roof Plumbing RP 02 | Box Gutters'

Audience

The audience/s for this Practice Note include/s:

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| ☒ Architects/ Designers | ☒ Owner Builders |
| ☒ Builders | ☒ Plumbers |
| ☒ Building Surveyors/ Inspectors | ☐ Real estate management agents |
| ☒ Engineers | ☒ Trades and Maintenance (inc. Electricians) |
| ☐ Homeowners / Residential Tenants | |

Purpose

This Practice Note provides guidance on the Deemed-to-Satisfy (DtS) requirements for the installation of box gutters with sole widths between 200mm and 600mm.

The content below provides guidance on:

- Defining a box gutter
- Design and installation parameters for box gutters
- Requirements for a box gutter support system
- Overflow provisions in a box gutter

Abbreviations & Definitions

The abbreviations and definitions set out below are for guidance only. They are not intended to vary those set out in the Building Act 1993, the Plumbing Regulations 2018 or the National Construction Code.

- **Act** – Building Act 1993
- **NCC** – National Construction Code 2022
- **AS** – Australian Standard
- **AS/NZS** - Australian/New Zealand Standard
- **DtS** – Deemed-to-Satisfy
- **PCA**– Plumbing Code of Australia (National Construction Code 2022, Volume Three)

Defining a box gutter

A box gutter is defined as a graded channel, generally of rectangular shape, for the conveyance of rainwater within the building footprint, typically adjacent to a wall or parapet. A box gutter incorporating a lear is also an acceptable shape, provided the minimum sole widths are observed and the effective cross-sectional area of the gutter is appropriately sized for the roof catchment area.

All box gutter installations, whether new or replacement must satisfy the Performance Requirements of the Plumbing Code of Australia (PCA). The Performance Requirements apply to the design, construction, installation, replacement, repair, alteration and maintenance of box gutters. Compliance with the Performance Requirements of the PCA are automatically satisfied by complying with the AS/NZS 3500.3 and HB 39. Alternatively, they can be demonstrated through the Performance Solution process as set out in Section 2 of the PCA.

NB: V-shaped gutters are not permitted. If installed, they may result in water damage to the building. Damage may be caused by premature failure of the gutter due to inadequate drainage, permanent ponding and debris accumulating in the crevices, which may lead to intense localised corrosion of the gutter.

Deemed-to-Satisfy design and installation parameters for box gutters

The following information outlines the design parameters that must be adhered to for a box gutter installation to be considered a Deemed-to-Satisfy (DtS) installation:

- All box gutter sizing is based on an AEP (annual exceedance probability) of 1% (100 years ARI),
- The depth of box gutters and sizing of sumps, rainheads, downpipes and overflows must be designed using the general methods specified in the AS/NZS 3500.3,
- The maximum design flow per downpipe can only be plotted between 3 and 16 Litres per second,
- 100mm x 50mm downpipes are not an option that can be plotted from the standard for use with a sump,
- Box gutters sole widths can be plotted in 200mm, 300mm, 375mm 450mm, 525mm and 600mm. (can be a size in between if it is sized to the lower sole width),
- The minimum width of any box gutter: Domestic 200 mm and Commercial 300 mm,
- Box gutter sole widths of 200 mm cannot accommodate flow rates exceeding 10.5 Litres per second,
- Grade of box gutters can be plotted at 1:200, 1:150, 1:100 and 1:40,
- The depth of a sump with a high-capacity overflow must be sized using the general method and no sump/high-capacity combination shall be less than 150 mm in depth,
- Length of a sump with a high-capacity overflow is always 600mm,
- Length of a sump with a side overflow device shall not be less than 400mm,
- The width of any sump shall be equal to the width of the box gutter,
- Rainheads shall be left open above the overflow weir, inverted pops, Ned Kelly slots, round holes and vertical chutes or ducts are not deemed-to-satisfy solutions,
- Overflow devices must discharge to the atmosphere and be clear of neighbouring properties and public areas,
- Box gutters must be straight without a change of direction and discharge at the downstream end without a change in direction (i.e. not to the side),
- The box gutter sole width must not be reduced towards the outlet without a proportional increase in depth, the width of the gutter must not reduce to less than the minimum width at which it was designed (i.e. if designed at 200 mm sole width, gutter must not reduce to less than 200mm in width),
- Sumps and rainheads must be fixed and fully sealed to the box gutter,
- All box gutters must incorporate provision for expansion; where the distance between fixed points exceeds 6 metres; and at appropriate intervals for the material and situation as prescribed by the standard,
- No part of the outlet is above the sole of the sump or rainhead, and
- Lap joints of box gutters to have 25mm laps sealed and fastened in the direction of fall.

NOTE: some requirements of HB 39 conflict with the requirements of AS/NZS 3500.3, in these situations the requirements of AS/NZS 3500.3 shall be used.

Deemed-to-Satisfy requirements for box gutter overflow provision.

Overflow devices are critical to a box gutter installation. Failing to install appropriately sized and positioned overflow devices can lead to serious damage to buildings and contents, often resulting in high-cost insurance claims.

The Plumbing Code of Australia (PCA) Vic Part E3 sets out the Performance Requirements to safeguard people from illness, injury or loss (including loss of amenity) due to the failure of a roof drainage installation.

Compliance with these requirements can be demonstrated through either a Performance Solution or a deemed-to-satisfy solution, however, roof drainage systems that are designed and installed in accordance with AS/NZS 3500.3 and SA HB:39 are automatically deemed-to-satisfy these requirements.

Where a Performance Solution is used as the compliance pathway, the solution must demonstrate that the box gutter's functionality is at least equivalent to its deemed-to-satisfy alternative. This will ensure that the box gutter, downpipe, and overflow sizing has sufficient hydraulic capacity and freeboard to prevent wind driven spillages and overtopping from occurring within the building's footprint in the event of a total downpipe blockage.

Rainhead overflow provisions: To ensure that adequate overflow provision is made, and any surcharge is accommodated, the overflow weir of the rainhead must be the full width of the rainhead with the height of the weir positioned 25mm below the box gutter sole and be fully open above the weir at the front of the rainhead.

Sumps and overflow provisions: Box gutter sumps must incorporate either a side duct/channel overflow, or a high-capacity overflow device. A side overflow device is typically associated with an internal box gutter alongside a parapet wall. In the event of total or partial blockage of outlets or downpipes, the overflow device must discharge to atmosphere and remain within the property that it serves. The location of the discharge should be chosen so as to not cause a nuisance or damage to the property.

The size of the overflow device must be calculated in the accordance with the general methods prescribed by the AS/NZS 3500.3.

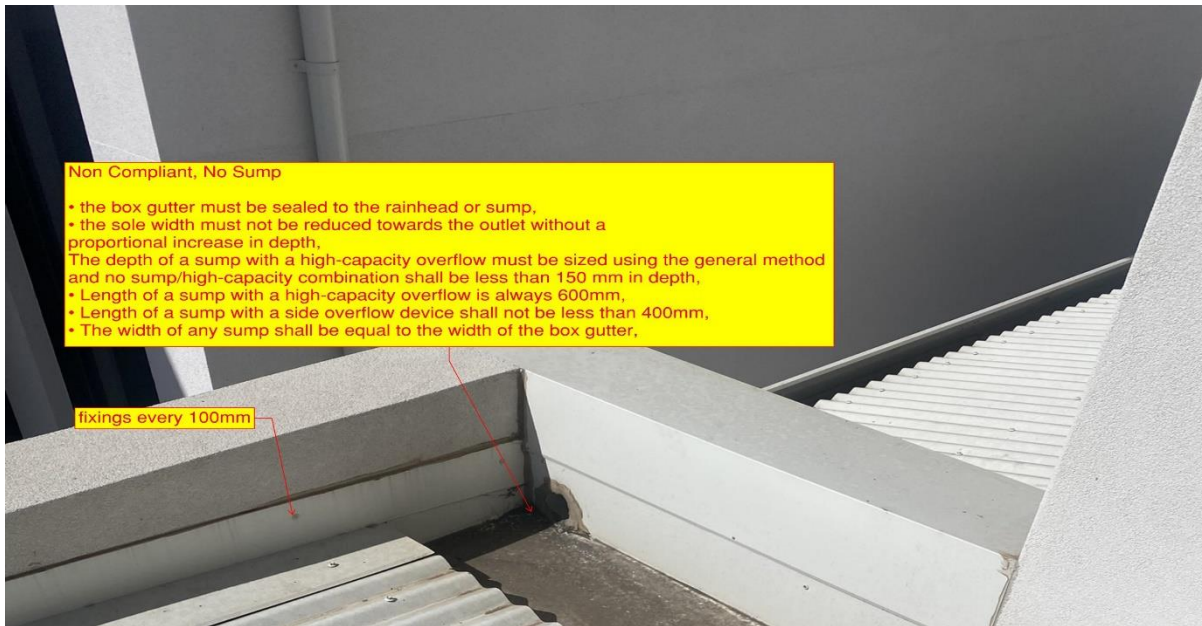


Figure 1: Example of a non-compliant box gutter.

Collaboration on the design of box gutters

Designers, architects and building surveyors must ensure that approved designs can satisfy the Performance Requirements of the PCA. This means where design parameters of AS/NZS 3500.3 and HB 39 cannot be adhered to a Performance Solution must be agreed between the relevant stakeholders prior to work commencing. During construction, if designs are found to be outside of the DtS parameters, the roof plumber should consult with the Relevant Building Surveyor and builder as a Performance Solution may need to be developed and an amended Building Permit may need to be issued.

Related Documentation

- Building Act 1993
- Building Regulations 2018
- National Construction Code, Volume 3, Plumbing Code of Australia (PCA) 2022: VIC Part E3
- AS/NZS 3500.3:2021 Part 3: Stormwater Drainage
- HB 39:2015 Amd 1:2021 Installation code for metal roofing and cladding
- Plumbing Practice Note RP-01: Regulatory Framework
- Plumbing Practice Note RP-03: Eaves Gutters
- Plumbing Practice Note RP-04: Downpipes
- Plumbing Practice Note RP-05: Flashings

